Curl and Divergence

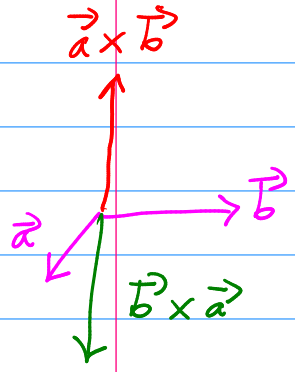
1) Definitions

Defn: The curl of a vector field $\vec{F}$ is the cross product $\vec{\nabla} \times \vec{F}$. (curl $(\vec{F})$)
The divergence of $\vec{F}$ is the scalar/dot product $\vec{\nabla} \cdot \vec{F}$. (div $(\vec{F})$)

Remark: curl $(\vec{F})$ is defined for "3D" $\vec{F}$.

- $\vec{\nabla} = \left\langle \frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right\rangle$

- $\vec{a} = \langle a_1, a_2, a_3 \rangle$ and $\vec{b} = \langle b_1, b_2, b_3 \rangle$



$$\vec{a} \times \vec{b} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

$$= (a_2 b_3 - b_2 a_3)\vec{i} + (a_1 b_3 - b_1 a_3)\vec{j} + (a_1 b_2 - b_1 a_2)\vec{k}$$

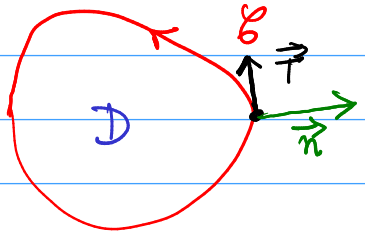
($\vec{i} \times \vec{j} = \vec{k}$, $\vec{j} \times \vec{k} = \vec{i}$, $\vec{k} \times \vec{i} = \vec{j}$)

$$\Rightarrow \vec{\nabla} \times \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ P & Q & R \end{vmatrix}$$

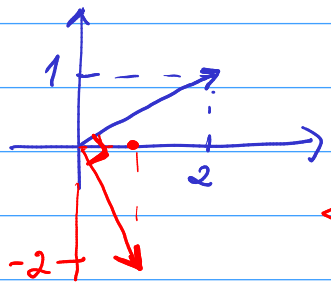
Vector form of Green's Theorem:

$$\oint_{\mathcal{C}} \vec{F} \cdot d\vec{r} = \iint_D (\operatorname{curl}(\vec{F}) \cdot \vec{k})\, dA$$

- Line integral $\vec{F}$ along the normal $\vec{n}$



$\langle a, b \rangle \longmapsto \langle b, -a \rangle$



$\langle 2, 1 \rangle$

$\langle 1, -2 \rangle$

$$\vec{T} = \frac{\vec{r}'(t)}{|\vec{r}'(t)|} = \left\langle \frac{x'(t)}{|\vec{r}'(t)|}, \frac{y'(t)}{|\vec{r}'(t)|} \right\rangle$$

$$\Rightarrow \quad \vec{n} = \left\langle \frac{y'(t)}{|\vec{r}'(t)|}, -\frac{x'(t)}{|\vec{r}'(t)|} \right\rangle$$

$d\vec{r} = \langle dx, dy \rangle$ $\qquad (\vec{T} \cdot ds)$

$\vec{n} \cdot ds = \langle dy, -dx \rangle$

$$\Rightarrow \oint_{\mathcal{C}} \vec{F} \cdot \vec{n}\, ds = \oint_{\mathcal{C}} \underbrace{P\,dy - Q\,dx}$$

$\langle P, Q \rangle \cdot \langle dy, -dx \rangle$

We have
$$\oint_{\mathcal{C}} P\,dy - Q\,dx = \iint_D \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} \right) dA$$
$$= \iint_D \operatorname{div}(\vec{F})\, dA$$

Proof (sketch) $\oint_{\mathcal{C}} P\,dy - Q\,dx = ?$

$A = -Q$

$B = P$

$$\oint_C P\,dy - Q\,dx = \oint_C A\,dx + B\,dy$$

$$= \iint_D \left(\frac{\partial B}{\partial x} - \frac{\partial A}{\partial y}\right) dA$$

$$= \iint_D \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y}\right) dA$$

$$= \iint_D \operatorname{div}(\vec{F})\, dA$$

